Nagpur Real Estate Platform — Development Documentation

Website · Admin Dashboard · Mobile App (Android + iOS) · Working name: [Platform Name – TBD] · Draft v1 · 6 Nov 2026

1. Purpose and Scope

This document tells the engineering team what to build and how. It covers **three deliverables on one shared backend**:

#	Deliverable	Users	Goal
1	**Public website** (basic, SEO-led)	Buyers, renters, owners	Search, view and enquire about properties in Nagpur, in English, Hindi and Marathi
2	**Mobile app** (Android + iOS)	Same as the website, plus repeat users	Same features as the website, optimised for one-hand use, OTP login and push alerts
3	**Admin dashboard**	Admin, moderators, agents, owners	Verify listings, manage leads (CRM), content, users and analytics

All three use **one API and one database**, so a listing posted on the app appears on the website and in the dashboard instantly.

Out of scope for Phase 1: in-app real-time chat, price-trend graphs, virtual tours, interiors and legal service modules, paid subscriptions. These are listed in Phase 2 (section 14).

Changes from the Initial Plan

- **Mobile app:** Flutter replaces React Native (Expo). Flutter gives one codebase, consistent rendering on low-end Android phones and strong map and list performance.
- **Database:** PostgreSQL + PostGIS replaces MongoDB. Property data is relational (listings, localities, agents, leads) and map and radius search is a core feature.
- **Backend:** NestJS (TypeScript) replaces plain Express, for structure, validation and auto-generated API docs.

2. Reference Study: What We Take from Zameen.com

Zameen.com is Pakistan's largest property portal. We use it as a **structural reference only**; the brand, content, imagery and code are our own and Nagpur-specific.

Zameen feature	Our decision
Home search bar: Buy/Rent toggle, city, location, property type, area, beds, price	**Adopt.** City is fixed to Nagpur, so the first field is locality. Area in sq ft (with sq m option).
Top navigation: Buy, Rent, New Projects, Agents, More	**Adopt.** Buy, Rent, Plots, Commercial, Projects, Agents, Tools.
Property type split: Homes, Plots, Commercial	**Adopt** with Flats, Houses, Plots, Shops/Offices as sub-types.
Popular-locations link blocks on the home page (per type and city)	**Adopt.** Excellent for SEO. We generate them from live listing counts per Nagpur locality.
New Projects (builder projects with units)	**Adopt** in Phase 1 as a simple Projects module.
Agents directory and agency pages	**Adopt.** Verified agent profiles with active listings.
Tools: home loan calculator, area unit converter, construction cost calculator, plot finder	**Adopt** the loan/EMI calculator and area converter in Phase 1. Construction cost calculator in Phase 2.
Area guides, property index and trends	**Adopt** area guides in Phase 1 (CMS pages). Price index and trends in Phase 2.
Profolio (agent dashboard to manage all listings)	**Adopt** as the Agent/Owner view of our admin dashboard.
Add Property flow	**Adopt** with phone OTP and a short stepper.
Z-Value (property valuation)	**Phase 3 idea.** Not in scope.
Forum, expo, investment property blocks	**Skip.**
Download-the-app prompts	**Adopt.** Smart banner on mobile web.

Where we differ: a lighter, brighter and simpler UI, local languages from day one, "verified" and RERA badges on every card, EMI shown on every property, and Call / WhatsApp always one tap away.

3. Recommended Tech Stack

3.1 Summary

Layer	Choice	Why
Website	Next.js (App Router, TypeScript) + Tailwind CSS + next-intl	Server rendering and ISR for SEO, fast pages, locale-prefixed routes for English/Hindi/Marathi
Admin dashboard	Next.js (client-heavy) + shadcn/ui + TanStack Table and Query + React Hook Form + Zod + Recharts	Same language and tooling as the website; strong table, form and chart libraries
Mobile app	Flutter (Dart 3)	One codebase for Android and iOS, smooth lists and maps, good low-end device performance
API	NestJS (Node.js, TypeScript), REST + OpenAPI	Modular, typed, auto-generated docs and client SDKs for web and Flutter
Database	PostgreSQL + PostGIS	Relational data, geo queries (radius, polygon, nearest metro/school)
ORM	Prisma (typed raw SQL for PostGIS queries)	Safe migrations and productivity
Search	Meilisearch (self-hosted)	Typo tolerance and Devanagari support, instant filtering and facets
Cache / queues	Redis + BullMQ	Rate limits, OTP throttling, background jobs (emails, notifications, image processing)
Media	AWS S3 + CloudFront + image-resize Lambda (WebP/AVIF, watermark)	Fast images on slow networks
Maps	Google Maps Platform (Maps, Places, Geocoding)	Most familiar to local users; set daily quotas and budget alerts
Auth	Phone OTP (public, agent, owner); email + password + TOTP 2FA (admin staff). JWT access + rotating refresh tokens	No passwords for the public; strong control for staff

Layer	Choice	Why
SMS / WhatsApp	MSG91 (DLT-compliant SMS and OTP), WhatsApp Business Cloud API	Indian SMS needs DLT registration; WhatsApp matches user habits
Push	Firebase Cloud Messaging (Android and iOS via APNs)	Standard and free
Email	AWS SES	Low cost transactional mail
CMS	Sanity.io (blog, area guides, FAQ) or built-in admin pages	Editors work without developers
Analytics / errors	GA4 + PostHog (product), Sentry (web, API, Flutter), Firebase Crashlytics (app)	Funnel and crash visibility
Hosting	AWS Mumbai (ap-south-1): ECS Fargate, RDS PostgreSQL, ElastiCache Redis	Low latency for Nagpur users and data stays in India
CI/CD	GitHub Actions + Docker; Fastlane for app builds; Terraform for infra	Repeatable releases
Repo	Monorepo: pnpm + Turborepo (web, dashboard, api, shared packages) and the Flutter app in the same repo	Shared types and one pull-request flow

3.2 Key Decisions and Alternatives

- **Why not one framework for web and app?** SEO needs server rendering (Next.js), while the app needs native performance and offline caching (Flutter). Both read the same OpenAPI contract, so features stay identical.
- **Why not Firebase as the whole backend?** Property search with geo filters, lead routing and moderation workflows fit a relational database and a proper API better.
- **Cheaper start:** the web and dashboard can run on Vercel (Mumbai region) while the API stays on AWS. We recommend all-AWS for simpler operations and cost control.
- **Search scale-up path:** Meilisearch is enough for tens of thousands of listings. Move to OpenSearch only if volume or analytics needs grow far beyond Nagpur.

4. System Architecture

Request flow: Website, Dashboard and Flutter App → CloudFront / Load Balancer → NestJS API → PostgreSQL (source of truth), Meilisearch (search index), Redis (cache, queues), S3 (media). A background worker consumes BullMQ jobs for notifications, image processing, search indexing and lead routing.

4.1 Repository Layout

```
/apps
  /web            Next.js public website
  /dashboard      Next.js admin dashboard
  /api            NestJS API + worker
  /mobile         Flutter app (Android + iOS)
/packages
  /api-client     TypeScript client generated from OpenAPI
  /ui             Shared design tokens and components (web + dashboard)
  /config         ESLint, TypeScript, Tailwind presets
/infra            Terraform, Dockerfiles, GitHub Actions
/docs             This document, ADRs, API changelog
```

The Flutter app uses a Dart client generated from the same OpenAPI file in CI, so any API change that breaks the app fails the build.

4.2 API Conventions

- REST, versioned under `/v1`, JSON only, OpenAPI 3 spec published at `/docs`.
- Cursor-based pagination, `Accept-Language` header for en/hi/mr content.
- Consistent error shape: `{ code, message, details }`.
- Rate limits per IP and per phone number (strict on OTP: for example 5 per hour per number).
- Idempotency keys on enquiry and listing submission to avoid duplicates on slow networks.

5. Data Model (Core Entities)

Entity	Key fields
User	id, phone (unique), name, email, role (public, agent, owner, moderator, admin), language, status, last_login

Entity	Key fields
AgentProfile	user_id, agency name, RERA agent number, bio, areas served, verified, rating
Property	id, slug, purpose (buy/rent), type (flat, house, plot, shop, office), title, description, price, price_unit, area_sqft, bedrooms, bathrooms, furnishing, facing, floor, age, possession_status, possession_date, rera_number, locality_id, project_id, geo point (PostGIS), owner_id, status, verified, featured, expires_at
PropertyMedia	property_id, type (photo, video, floor plan, brochure), url, order, alt text
Locality	id, name (en/hi/mr), slug, polygon, centroid, pincode, guide content, listing counts
Project / Builder	name, builder, RERA number, status, units, amenities, location, brochure
Amenity / PropertyAmenity	master list and mapping
Enquiry (Lead)	id, property_id, user_id or guest phone, source (web, app, WhatsApp), message, status, assigned_agent_id, SLA due time
LeadActivity	lead_id, actor, type (call, note, status change), notes, timestamp
SiteVisit	lead_id, property_id, slot, status (requested, confirmed, done, cancelled)
SavedProperty / SavedSearch	user_id, property_id or filter JSON, alert frequency
Verification	property_id, checks (owner identity, ownership document, RERA, photo match), reviewer, outcome, remarks
Device	user_id, FCM token, platform, app version
Notification	user_id, channel, template, payload, status
ContentPage	type (blog, area guide, FAQ), locale versions, SEO fields
LoanRequest	user_id, property_id, price, down payment, tenure, status, partner

Entity	Key fields
AuditLog	actor, action, entity, before/after, IP, timestamp

5.1 Listing Lifecycle

`draft` → `pending_review` → `live` → `expired` / `sold_or_rented` / `archived`, with `rejected` (reason sent to the owner) as a side exit from review. Live listings expire after 90 days unless renewed, and the owner gets reminders at 7 and 1 day before expiry.

6. API Surface (Key Endpoint Groups)

Group	Examples
Auth	`POST /v1/auth/otp/request`, `POST /v1/auth/otp/verify`, `POST /v1/auth/refresh`, `POST /v1/auth/logout`
Search	`GET /v1/properties` (filters: purpose, type, locality, price range, beds, possession, verified, bbox, near lat/lng, sort), `GET /v1/properties/suggest` (autocomplete)
Property	`GET /v1/properties/{slug}`, `GET /v1/properties/{id}/similar`, `POST /v1/properties` (create), `PATCH /v1/properties/{id}`, `POST /v1/properties/{id}/media/presign`
Enquiry	`POST /v1/enquiries`, `GET /v1/me/enquiries`, `POST /v1/site-visits`
User	`GET/PATCH /v1/me`, `GET/POST/DELETE /v1/me/saved-properties`, `GET/POST /v1/me/saved-searches`, `POST /v1/me/devices`
Localities	`GET /v1/localities`, `GET /v1/localities/{slug}` (guide, counts, nearby places)
Projects and agents	`GET /v1/projects`, `GET /v1/agents`, `GET /v1/agents/{id}`
Tools	`GET /v1/tools/emi`, `GET /v1/tools/area-convert` (also available client-side)

Group	Examples
Content	`GET /v1/content/{type}`
Admin	`GET /v1/admin/listings?status=pending`, `POST /v1/admin/listings/{id}/approve`

7. Website Specification

7.1 Routes

Route	Purpose	Rendering
`/[locale]`	Home	ISR, revalidate every few minutes
`/[locale]/buy`, `/rent`, `/plots`, `/commercial`	Listing pages with filters, grid / list / map toggle	SSR with filter state in the URL
`/[locale]/nagpur/[locality]/[type]-for-[purpose]`	Programmatic SEO pages (for example Dharampeth flats for sale)	ISR
`/[locale]/property/[slug]`	Property detail	ISR with on-demand revalidation when the listing changes
`/[locale]/localities`, `/localities/[slug]`	Locality guides	ISR
`/[locale]/projects`, `/projects/[slug]`	New projects	ISR
`/[locale]/agents`, `/agents/[id]`	Agent directory and profile	ISR
`/[locale]/post-property`	Post property stepper	Client, login required at submit
`/[locale]/tools/emi-calculator`, `/tools/area-converter`	Tools	Static
`/[locale]/blog`, `/blog/[slug]`, `/about`, `/contact`, `/faq`	Content	Static / ISR

Route	Purpose	Rendering
`/[locale]/account/*`	Saved, enquiries, my listings	Client, authenticated

7.2 Homepage Sections

Header (logo, language switch, Post Property, Login) → search block (Buy, Rent, Plots, Commercial tabs, locality autocomplete, budget, bedrooms, Search) → quick chips (1 BHK, 2 BHK, 3 BHK, Plots, Shops, Ready to Move) → popular localities → verified and featured properties → how it works → EMI calculator → services → testimonials → enquiry form and WhatsApp button → footer with contact and RERA information.

7.3 Property Card and Detail Rules

- **Card:** photo, price, EMI per month, size, locality, Verified tag, posted date, Call and WhatsApp buttons.
- **Detail:** gallery, price and EMI at the top, key facts, map with nearby schools, hospitals, metro and markets, floor plan and brochure, similar properties, and a sticky Call · WhatsApp · Book Visit bar.
- Call and WhatsApp work without login. Login is requested only to save, book a visit or post.

7.4 SEO and Performance Requirements

- Per-page title, description, canonical, `hreflang` for en/hi/mr, Open Graph tags.
- JSON-LD: `RealEstateListing`, `Organization`, `BreadcrumbList`, `FAQPage`. Auto-generated XML sitemaps, split by type and locale.
- Clean English slugs; translated titles and descriptions.
- Core Web Vitals targets on mid-range phones over 4G: LCP under 2.5 s, INP under 200 ms, CLS under 0.1.
- Images: responsive `srcset`, WebP/AVIF, lazy loading, skeleton loaders. Map loads only when its toggle or section is opened.
- Accessibility: WCAG 2.1 AA, 48 px minimum touch targets, visible focus, correct Devanagari fonts (Noto Sans Devanagari).
- Mobile web shows a smart banner linking to the app.

7.5 Internationalisation

All UI strings live in message files (`en.json`, `hi.json`, `mr.json`). Prices use the Indian numbering format (lakh and crore). Dates and numbers follow the selected locale. User-written listing text stays in the language it was written in; auto-translation is a Phase 2 option.

8. Mobile App Specification (Flutter)

8.1 Architecture

- **Pattern:** feature-first folders with clean layers (presentation, domain, data).
- **State management:** Riverpod. **Navigation:** go_router with deep links.
- **Networking:** dio with a generated OpenAPI client, token refresh interceptor, retry with backoff.
- **Local storage:** drift (SQLite) for cached listings and saved properties; flutter_secure_storage for tokens.
- **Models:** freezed + json_serializable.
- **Key packages:** google_maps_flutter, geolocator, cached_network_image, firebase_messaging, flutter_local_notifications, url_launcher (call, WhatsApp), local_auth (biometric unlock for agent and owner accounts), intl and Flutter's ARB localisation, share_plus, sentry_flutter, firebase_crashlytics.
- **Environments:** dev, staging, prod flavours with separate Firebase projects and API URLs.

8.2 Navigation and Screens

Bottom navigation: **Home · Explore · Post · Saved · Account**.

Flow: Language select → Home → Search results → Filters (bottom sheet) → Property detail → Enquiry or Site visit → Saved → Post property → Account (profile, my enquiries, my listings, language, help).

8.3 App-Specific Behaviour

- **Login:** phone number and OTP, with SMS auto-read on Android and an iOS one-time-code field. Guests can browse and call.
- **Push:** new listing matching a saved search, price drop, enquiry status update, visit reminder. Ask for notification permission after the first save, not on launch.
- **Offline:** last viewed and saved properties are readable without network; images cached; an offline banner and retry on search.

- **Near me:** location permission requested only when the user taps Near me.
- **Deep links:** Android App Links and iOS Universal Links so shared website URLs open the matching screen.
- **Performance targets:** cold start under 3 s on a mid-range Android phone, list scrolling at 60 fps, app size under 40 MB.
- **Support:** Android 8.0 (API 26) and above, iOS 14 and above.

8.4 Release

Fastlane lanes for internal testing (Firebase App Distribution), TestFlight and Play Console tracks. Staged rollout (10%, 50%, 100%) with crash-free target of 99.5% or higher. Store listings in English, Hindi and Marathi. Privacy labels and the Data Safety form must be completed before submission.

9. Admin Dashboard Specification

9.1 Roles and Permissions (RBAC)

Role	Access
Super Admin	Everything, including staff management, settings and audit logs
Moderator	Review, approve and reject listings; manage reports and content
Sales / CRM Manager	All leads, assignment rules, agent performance
Agent	Own listings, own leads, own profile and enquiry history
Owner / Seller	Own listings and their enquiries (simplified view)
Content Editor	Blog, area guides, FAQ, banners

Permissions are enforced on the API, not only hidden in the UI. Every admin action writes to the audit log.

9.2 Modules

Module	Features
Overview	Today’s leads, new listings, pending reviews, top localities, conversion funnel
Listings	Table with filters and bulk actions; create/edit with photo upload and map pin; status control, feature or unfeature, expiry and renewal
Moderation queue	Pending listings with a checklist (owner identity, ownership document, RERA number, photo quality, duplicate detection), approve or reject with reason, SLA timer
Leads (CRM)	Kanban and table views: New → Contacted → Visit Scheduled → Closed (won/lost); assignment, notes, call log, follow-up reminders, WhatsApp click-to-chat, CSV export
Lead routing	Rules by locality, property type and agent availability; round-robin fallback; escalation if untouched past the SLA
Site visits	Calendar of requested and confirmed visits
Users and agents	Search, verify agent profiles, suspend or reactivate, view activity
Projects and builders	Manage projects, units, brochures and RERA details
Localities	Edit locality details, guide content, polygons and featured status
Content	Blog, area guides, FAQ, banners and homepage featured blocks, in en/hi/mr
Notifications	Send push or WhatsApp broadcasts, manage templates
Loan requests	Requests from the EMI tool, status and partner hand-off
Analytics	Traffic, searches with no results, listing views, enquiry-to-visit conversion, agent response time, exports

Module	Features
Settings	Platform settings, plans and expiry rules, SEO defaults, staff and roles, audit log viewer

9.3 UX Rules

Desktop-first, usable on tablet. Server-side pagination, sorting and filtering for every table. Saved filter views. Autosave on long forms. Keyboard shortcuts for approve and reject in the moderation queue. Dashboard login with email, password and TOTP two-factor.

10. Cross-Cutting Systems

10.1 Search

PostgreSQL is the source of truth. On every listing change a job updates the Meilisearch index (properties, localities, projects, agents). Searchable fields include title, locality names in all three languages and transliterations (for example “Dharampeth / धरमपेठ”). Filters and facets come from the index, while map bounding-box and radius queries use PostGIS. Search queries with zero results are logged for the team to act on.

10.2 Lead Handling

Enquiry created → saved with idempotency key → routing rule picks an agent → push, WhatsApp and email notification to the agent within a minute → customer receives a confirmation → agent updates the status → unattended leads escalate after the SLA. All steps are visible in the lead timeline.

10.3 Media Pipeline

Client requests a pre-signed S3 URL → uploads directly → a worker generates thumbnails, WebP/AVIF variants and a watermark → the original is kept privately. Limits: 20 photos and one video per listing, 10 MB per photo. Basic checks for blur, minimum resolution and duplicates.

10.3 Notifications

One notification service with templates per channel (push, SMS, WhatsApp, email) and per language. User preferences and quiet hours are respected. WhatsApp messages use pre-approved templates.

11. Security and Compliance

- **Transport and data:** HTTPS everywhere, encryption at rest (RDS, S3), secrets in AWS Secrets Manager, least-privilege IAM.
- **Auth:** short-lived access tokens, rotating refresh tokens, device binding for the app, OTP rate limiting and lockouts, TOTP for staff.
- **App security:** input validation with Zod / class-validator, output encoding, CSP headers, CORS allow-list, file type and size checks, signed URLs for private documents.
- **Abuse control:** CAPTCHA on guest enquiries and OTP requests, duplicate and spam listing detection, report-listing button.
- **Privacy (India's DPDP Act, 2023):** explicit consent at sign-up and enquiry, a clear privacy policy, data access and deletion requests handled from the account screen, minimal data collected. Do not store Aadhaar numbers; if identity checks are needed, use a compliant verification provider and store only the result.
- **RERA:** show the MahaRERA registration number on projects and applicable listings, with a link to the official record; moderators verify it.
- **Backups:** daily automated database backups with point-in-time recovery, tested restores each quarter.
- **Pre-launch:** third-party penetration test or at least an OWASP Top 10 review.

12. DevOps and Quality

12.1 Environments and CI/CD

Environments: local (Docker Compose), staging, production. Every pull request runs lint, type check, unit tests and a preview build; merging to `main` deploys to staging automatically; production deploys need manual approval. Infrastructure is defined in Terraform. Database migrations run through Prisma and are reviewed in pull requests. Feature flags are used for Phase 2 features.

12.2 Monitoring

Sentry for web, API and app errors; CloudWatch dashboards and alarms (error rate, latency, queue depth, database CPU); uptime checks on web and API; weekly budget review of Google Maps, SMS and WhatsApp usage.

12.3 Testing Strategy

Level	Tooling	Scope
Unit	Jest (API, web), flutter_test	Business rules, EMI math, filters, routing rules
API / integration	Supertest + test database	Auth, search, listing lifecycle, enquiry routing
End-to-end	Playwright (web and dashboard), Flutter integration tests	Search → detail → enquiry, post property, moderation
Performance	Lighthouse CI, k6	Web vitals budgets, search API under load
Accessibility	axe-core	Key pages and flows
Device matrix	Real devices	Low-end Android (2–3 GB RAM), mid-range Android, two iPhones
UAT	Real Nagpur users	5 to 10 buyers, renters and owners before launch

Definition of done: feature meets the acceptance criteria, has tests, is translated in all three languages, passes accessibility and performance budgets, and is documented in the OpenAPI spec or changelog.

13. Delivery Plan (14 Weeks, Seven 2-Week Sprints)

Sprint	Weeks	Focus	Output
0	1–3 (overlaps)	Discovery and UX	Wireframes and UI design in Figma, design tokens, tested with 5–10 users; repo, CI and infrastructure set up
1	3–4	Foundations	Auth (OTP), user and role model, locality and property schema, design system in web and Flutter
2	5–6	Search and listings	Property CRUD, media pipeline, Meilisearch indexing, listing page and property detail on web

Sprint	Weeks	Focus	Output
3	7–8	Web completion and app core	Home, localities, projects, agents, EMI tool, SEO pages; Flutter home, search, filters, detail
4	9–10	Leads and dashboard	Enquiry, site visit, lead routing, CRM, moderation queue, admin listings
5	11–12	App completion and integrations	Saved, post property, push, deep links, WhatsApp and SMS, content module, analytics
6	13–14	Hardening and launch	UAT, performance, security review, store submission, data migration, go-live and monitoring

Timeline and cost are finalised once the open questions below are answered.

14. Phase 2 Backlog

Real-time chat (Socket.IO or a managed service), saved-search alert digests, price trends and property index, commute-time filter, draw-on-map search, interiors and legal service modules, video and virtual tours, agent performance analytics, paid featured listings and subscriptions (Razorpay), construction cost calculator, auto-translation of listings.

15. Risks and Open Questions

Risk	Mitigation
Low listing supply at launch	Seed with the client's own inventory and partner builders; owner-friendly posting flow; moderation SLA of 24 hours
Fake or duplicate listings	Verification checklist, duplicate detection, report button, agent verification
SMS delivery and DLT approval delays	Start DLT registration in week 1; add WhatsApp OTP fallback
Google Maps cost overruns	Quotas, budget alerts, lazy map loading, cached geocodes
Weak connectivity	Image optimisation, skeleton loaders, offline cache in the app

Risk	Mitigation
Store rejection	Early privacy, data-safety and account-deletion compliance; submit a build by week 12

Questions to confirm before build starts:

1. Final platform name, logo and brand colours (the proposed direction is deep teal or navy with an orange accent).
2. Is the firm a broker, a builder or both, and may outside owners and agents post listings?
3. Property types in scope for launch: flats, houses, plots, commercial, rentals.
4. Languages required at launch (the plan assumes English, Hindi and Marathi).
5. Existing inventory, data format and any builder partnerships, for data import.
6. Loan, interiors and legal partners to integrate, and whether leads are shared with them.
7. Whether the earlier document you mentioned adds requirements beyond the initial plan; if so, share it and we will merge it into this document.
8. Budget range, target launch date, and who owns the Apple and Google developer accounts (they must be registered in the client’s name).